vllm-sr/vela3-lite

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Name it. Vela finds it.

vLLM Semantic Router × KR Labs · Vela 3.0 family · **vela3-lite** · 0.8B · base `d3-lite` · Apache-2.0

What it does · Quick start · Evaluation · The family · 📡 Collection

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vela3-lite is the 0.8B member of the Vela 3.0 family, built on d3-lite. In one `system_one` call it answers routing and general decisions and finds the exact spans you name (entities, relations, personal data,

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72

Inference Providers NEW

Zero-Shot Classification

This model isn't deployed by any Inference Provider.

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Model tree for `vllm-sr/vela3-lite`

Base model `Qwen/Qwen3.5-0.8B-Base`
Finetuned `Qwen/Qwen3.5-0.8B`
Finetuned `vllm-sr/d3-lite`
Adapter (6) this model

Collection including `vllm-sr/vela...`

📡 Vela 3.0 Collection
Six open routin... • 7 items • U... • △ 7

unsupported claims, evidence), with probabilities and character offsets and no generated text.

Parameters	0.85B with vision encoder (base) + 154 MB of adapters and heads
Base	vllm-sr/d3-lite at revision `9c107031`, unchanged
Inputs	Text or JSON, plus images
Answers	Choice · Yes / No · Score · Span
Licence	Apache-2.0

Summary

vela3-lite is the small d3 member of the Vela 3.0 family. d3-lite, frozen, answers every ordinary decision, bit for bit, and gated adapters add the spans you name, entities, relations and router signals. It has the family's cleanest router result, reads seven languages besides English for intents, and answers a span question in 29 ms at 200 tokens.

Highlights

- **Zero-shot NER:** 64.4 on OOD-7 (CrossNER + MIT), above every GLiNER / GLiFormer number reported on it (best: gliner-multitask-large-v0.5, 62.76; GNER-T5-large (780M): 63.5, reported).
- **Router signals:** better than Vela 2.0 0.8B on 23 of the 24 test files of the public router signal suite, level on 1, worse on none (paired 95% CI).
- **Multilingual intents:** on MASSIVE (test split, three unseen scenarios), intent accuracy in

German, French, Spanish, Italian, Polish, Chinese and Japanese is 76.4–83.7, within about 9 points of English (85.1); Hungarian (66.8) and Arabic (55.3) are partial.

- **Zero-shot classification:** 73.3 on the 13-set GLiFormer classification suite, above GLiFormer-base (264M): 72.36 and Qwen3.5-9B: 71.4 (reported, GLiFormer paper Tab 5).
- **Relations:** 19.7 end to end (GLiFormer Table 4 protocol, avg of CoNLL04 · DocRED · FewRel · CrossRE), above GLiFormer-base: 18.94 and GLiNER2: 17.8 (reported, GLiFormer paper Tab 4).
- **Decisions, bit for bit:** every ordinary decision is answered by `d3-lite@9c107031` itself, adapters off (561 of 561 identical in our check).



vela3-lite (0.8B) against reported numbers. Accent = vela3-lite; grey = other models (reported).

At a glance

What it is	A 0.85B decision-and-extraction model for LLM routing and guardrails. It runs vllm-sr/d3-lite, a small d3 decision model, unchanged, and adds the exact spans you name: entities,

	relations, personal data, unsupported claims and evidence.
How it was trained	d3-lite stays frozen, so every ordinary decision is d3-lite's own answer, bit for bit. Vela adds small gated LoRA adapters on top, one per job: spans (unsupported claims, personal data, evidence), entities, relations, router signals and image hallucination spans. Each adapter switches on only for its own questions. Router signals are trained supervised, as in Vela 2.0, on the public router datasets Vela 2.0 used (for example AEGIS 2.0, PolyGuardMix and Nemotron-Safety-Guard v3 for safety, Global-MMLU for domain, WildFeedback for feedback, DiffusionDB for modality, MASSIVE for intents); domain, feedback, hazard categories and fact-check needed also on the public router signal suite's train split. Other training data: public NER, relation, hallucination, evidence and PII datasets, plus web and document text labelled by a large teacher model. No test split and no Decision Index item is trained on; some router-signal-suite test rows also occur in public training sources, so the router table also reports the rows without that overlap.
What it is good at	Zero-shot NER; zero-shot relations; zero-shot classification; router signals above Vela 2.0 0.8B on 9 of 9 signals of the public router signal suite. Pick it for small GPUs and low latency.
How we measure it	Public test sets only, run through this package one request at a time with the shipped thresholds: CoNLL04, CoNLL04 · DocRED · FewRel · CrossRE (avg), OOD-7 (CrossNER + MIT), RAGTruth, Super-NI task 1568, the 13-set GLiFormer classification suite, the Decision Index 0.4.1, the LettuceDetect v2 test set, the public router signal suite. Other

	models' numbers are as their authors report them on the same test set and metric, or scored by us on the same rows where a table says so.

What it does

Capability	Ask (one question of a `system_one` call)	Get
Decisions	`{"type": "choice", "instructions": "Which topic?", "criteria": {"billing": None, "tech": None}}` (also `noul`, `score`)	a probability per option, from d3-lite itself
Router signals	a built-in router question id from `router_questions.json`	the signal's answer from the gated decision adapter
Named entities	`{"type": "span", "criteria": ["drug", "dosage"]}`	spans with label, offsets and score
Relations	`{"type": "span", "instructions": "Which spans are the founder of Apple?", "criteria": {"founder": None}}`	the related spans
Personal data	`{"type": "span", "criteria":`	spans of the 17 trained PII

Capability	Ask (one question of a system_one call)	Get
	`["PERSON", "EMAIL_ADDRESS"]}`	types, any subset
Unsupported claims	state `{"context": ..., "answer": ...}` and `{"type": "span", "criteria": {"unsupported": None}}`	the words of the answer the context does not support
Evidence	`{"type": "span", "criteria": ["evidence"]}`	the sentences that support the answer
Image-grounded claims	the unsupported-claims question with `images=["photo.jpg"]`	the words the image does not support

Quick start

```
# pip install "transformers==5.17.0" torch t
from transformers import AutoModel

m = AutoModel.from_pretrained("vllm-sr/vela3

# Name any labels; the spans come back with
r = m.system_one(
    state="Geoffrey Hinton presented capsule
    questions={"ents": {"type": "span", "cri
# researcher "Geoffrey Hinton", algorithm "c

# Unsupported claims in an answer, against a
r = m.system_one(
```

```
    state={"context": "The Eiffel Tower is 3
           "answer": "The Eiffel Tower is 330
    questions={"halu": {"type": "span", "crit
# flags "1925 by Gustave Eiffel's son"; add

# PII, relations and evidence use the same c
r = m.system_one(
    state="Steve Jobs founded Apple in Cuper
    questions={"pii": {"type": "span", "crite
               "rel": {"type": "span", "inst
```

Evaluation

All numbers on public test sets, measured through this package with `system_one`, one request at a time and the shipped thresholds. Other models' numbers are as reported by their authors (or scored in the same harness where marked). Full per-set tables: EVALUATION.md.

▼ Named entities

Benchmark	vela3-lite	Best other model (reported)	Vela 2.0 0.8B	Size-class best (0.5-1B, reported)
OOD-7, label names only (default; F1)	64.4	B2NER-20B 72.1 (standardised label names; B2NER Tab 3)	28.9	GNER-T5-large (780M) 63.5 (GNER, Table 2)
NER, 26 datasets of the GLiFormer paper (label	60.7	–	–	–

Benchmark	vela3-lite	Best other model (reported)	Vela 2.0 0.8B	Size-class best (0.5-1B, reported)
names; not all zero-shot)				

Label names are the only input. Training holds no CrossNER or MIT text, though many generic CrossNER type names occur as training labels.

▼ Relations

Benchmark	vela3-lite	Best other model (reported)	Vela 2.0 0.8B	Size-class best (0.5-1B, reported)
CoNLL04, head entity given, label names (F1)	**18.0**	GLiFormer-base 8.6 (same harness, Vela 2.0 card)	3.0	–
End to end, GLiFormer Table 4 protocol, avg of 4 (strict F1)	19.7	Qwen3.5-9B, prompted 25.7 (GLiFormer paper Tab 4)	–	GLiFormer-large (576M) 21.3
End to end: CoNLL04 · DocRED · FewRel · CrossRE	25.6 · 19.2 · 21.3 · 12.7	GPT-5-mini 42.4 · GLiNER-Relex 31.3 · Qwen3.5-	–	–

Benchmark	vela3-lite	Best other model (reported)	Vela 2.0 0.8B	Size-class best (0.5-1B, reported)
(CoNLL04 zero-shot, others in-domain)		9B 71.6 · GLiNER-Relex 18.1 (GLiFormer paper Tab 4)		

▼ Evidence

Benchmark	vela3-lite	Best other model (reported)	Vela 2.0 0.8B	Size-class best (0.5-1B, reported)
QASPER, word-F1 (zero-shot)	39.3	–	–	–
ACL-Verbatim, word-F1 (in-domain)	47.8	ACL-Verbatim ModernBERT 53.4 (in-domain, evidence only; Vela 2.0 card)	23.6	–

ACL-Verbatim is in-domain: its silver training split is in the training data.

▼ Unsupported claims (hallucination spans)

Benchmark	vela3-lite	Best other model (reported)	Vela 2.0 0.8B	Siz cla be (0.5- repor
LettuceDetect v2 test set, 10,698 answers (example-F1)	0.825	LettuceDetect v2 qwen-2b 0.921 (its card)	0.828	–
RAGTruth (example-F1)	0.675	HDM-2 (3B) 0.850 (HalluciNot Tab 2)	0.652	HDM- (0.5B) 0.789 (repo Hallu Tab 2
M-HalDetect validation, image-grounded (char-F1)	0.462	–	no image path	–

▼ Zero-shot classification

Benchmark	vela3-lite	Best other model (reported)	Vela 2.0 0.8B	Size-class best (0.5-1B, reported)
GLiFormer suite, 13 sets, label names only (mean macro-F1)	73.3	GPT-5-mini 79.8 (GLiFormer paper Tab 5)	–	GLiFormer-large (576M) 75.0

These questions run on d3-lite with every adapter off.

▼ **Router signals** (public router signal suite; above Vela 2.0 0.8B on 9 of 9)

Signal (metric, test rows)	vela3-lite	Vela 2.0 0.8B, same rows	Decision-1.0-Route-0.6B, same rows	Rows sharing no text with training: vela3-lite / Vela 2.0 0.8B
Harmful request (AUC, 2,230)	**0.931**	0.919	0.963	0.912 / 0.907 (1,254)
Jailbreak or prompt injection (AUC, 1,878)	**0.900**	0.872	0.991	**0.849** / 0.808 (944)
Personal data in the request (AUC, 3,000)	**0.944**	0.888	0.994	**0.942** / 0.885 (2,802)
Answer not supported by its context (AUC, 3,000)	**0.734**	0.716	0.791	**0.710** / 0.686 (1,743)
Fact-check needed (AUC, 800)	**0.940**	0.768	0.942	**0.931** / 0.774 (690)
Output modality (text or	**0.920**	0.841	0.999	**0.917** / 0.838 (2,916)

Signal (metric, test rows)	vela3-lite	Vela 2.0 0.8B, same rows	Decision-1.0-Route-0.6B, same rows	Rows sharing no text with training: vela3-lite / Vela 2.0 0.8B
image) (AUC, 3,000)				
Domain (subject area) (accuracy, 2,996)	**0.822**	0.703	0.863	**0.822** / 0.702 (2,916)
User feedback on the previous answer (accuracy, 2,842)	**0.808**	0.584	0.848	**0.789** / 0.543 (2,295)
Hazard categories (macro AUC, 11 labels, 3,001)	**0.919**	0.740	0.931	**0.884** / 0.702 (1,397)

vllm-sr/router-signal-suite (public), the `test` file of each signal, scored with the dataset's own scoring code. Each signal is a router question this package was trained on (supervised, as in Vela 2.0), asked one request at a time with the shipped thresholds.

Vela 2.0 0.8B was run by us through its public release on the same rows. Decision-1.0-Route-0.6B is scored

from the suite's published predictions; it was trained on the suite's own train split, so these test files are in-distribution for it.

Bold: the paired 95% bootstrap CI against Vela 2.0 excludes a tie. The last column drops the test rows that share text (exact or a 13-gram) with this model's training data. Safety and jailbreak are scored on the rows whose text the suite publishes (2,230 of 3,003 and 1,878 of 3,000).

Domain, feedback, hazard categories and fact-check needed are also trained on the suite's own train split (with the public router data Vela 2.0 used; the suite's test and hold rows, and anything sharing an exact text or a 13-gram with them, are kept out), as Decision-1.0-Route is, so their test files are in-distribution for both. The domain question has a 15th option, religion, which the suite maps to philosophy. Feedback has NO_FEEDBACK as a trained option. Hazard is a dedicated trained yes / no question per hazard category.

▼ **Decisions** (general decisions are the base model's, bit for bit)

Decision Index 0.4.1 (2026-10-28)	Public index	Full score (board rank)
Fastino GLiDE 1.2 (tied rank)	71.99	71.52 (1)
GLiClass v4 Edge (0.4B) (tied rank)	47.49	31.21 (156)
OneJev1.6-0.8B (tied rank)	48.76	29.58 (162)
vela3-fast (0.4B), the family's encoder (tied rank)	50.36	29.27 (164)

Decision Index 0.4.1 (2026-10-28)	Public index	Full score (board rank)
StartLux-Decision v1.1 0.8B (tied rank)	53.79	28.88 (167)
General decisions = d3-lite (this package's base)	36.93	26.16 (171)

General decisions are the base model's own answers, bit for bit, so the base's board entry is this package's decision score. Full score (the board's ranking): 60% private same-skill tests and 40% private new-domain tasks; public index: the 37 public benchmarks, chance-corrected. All values are board entries on the Decision Index 0.4.1 board (multimodalart/jev-decision-index, data generated 2026-10-28T09:02Z) unless noted; the rank is the position in the board's Full-score order, and tied rank marks entries the board shows as tied. On the board's Full score, d3-lite scores 26.16 (rank 171, tied).

Benchmark	vela3-lite	Best other model (reported)	Vela 2.0 0.8B	Size-class best (0.5-1B reported
Super-NI task 1568, held out (accuracy)	0.358	–	0.340	–
MASSIVE, three unseen scenarios: intent / scenario (accuracy; 14,911	0.601 / 0.707	–	0.544 / 0.632 (full set)	–

Benchmark	vela3-lite	Best other model (reported)	Vela 2.0 0.8B	Size-class best (0.5-1B reported)
decontaminated rows)				

MASSIVE: three scenarios never used in our router training (iot, transport, cooking), 52 locales; accuracy on the 14,911 test utterances that match no training text of the model's lineage. The 342 excluded utterances occur verbatim in upstream training mixtures (tasksource MASSIVE / HWU64 rows). Vela 2.0 values are on the full 15,253, so these rows are not bolded. Super-NI 1568: 79 to 90 of its 162 ProPara passages share a 13-gram with other tasks' training rows.

▼ **Multilingual intents** (MASSIVE test split, three unseen scenarios)

Language (MASSIVE locale)	Intent accuracy (%)	Scenario accuracy (%)
English (en-US)	85.1	93.0
German (de-DE)	78.8	88.7
French (fr-FR)	79.3	93.3
Spanish (es-ES)	79.1	92.1
Italian (it-IT)	78.8	92.8
Polish (pl-PL)	76.4	86.8
Chinese (zh-CN)	83.7	93.8
Japanese (ja-JP)	82.9	94.0

Language (MASSIVE locale)	Intent accuracy (%)	Scenario accuracy (%)
Hungarian (hu-HU), partial	66.8	83.9
Arabic (ar-SA), partial	55.3	73.6

MASSIVE test split, 416 utterances per locale (the same utterances in every locale), from the three scenarios our router training never used (cooking, iot, transport). All 60 intent and 18 scenario options are asked, with English option descriptions; one `system_one` call per row, shipped thresholds, nothing tuned for other languages. Majority class on the English rows: 17.3 intent / 52.9 scenario; 95% CI about ±2 to ±5 points at this size. This covers one task type (short assistant commands) with English labels; it does not measure the other signals (domain, personal data, safety, NER) in other languages.

▶ **Latency**

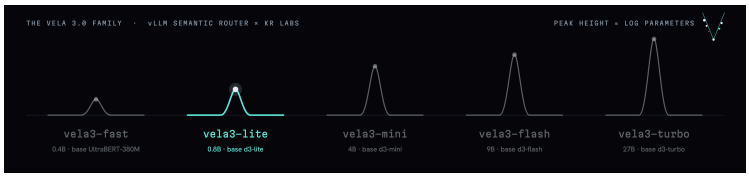
Benchmarks

- **OOD-7** (CrossNER AI, literature, music, politics, science; MIT movie, restaurant): find the entities of the named types in text from domains the model was not trained on (strict entity F1).
- **NER, 26 datasets of the GLiFormer paper** (IE_INSTRUCTIONS): entities of named types across 26 public NER sets, not all zero-shot (strict entity F1).

- **CoNLL04, head entity given:** given one entity of a sentence, find the entities it is related to by the named relation (F1 over those tail entities).
- **Relations end to end, GLiFormer Table 4 protocol** (CoNLL04, DocRED, FewRel, CrossRE): find the entity pairs and their relation from raw text (strict triplet F1).
- **QASPER:** find the sentences of a research paper that answer a question about it (word-F1).
- **ACL-Verbatim** (KRLabsOrg/acl-verbatim-spans): find the passage of an ACL paper that supports an answer (word-F1).
- **M-HalDetect** (validation): which words of an image description the image does not support (char-F1).
- **LettuceDetect v2 test set** (10,698 answers, as in the LettuceDetect v2 paper (GroundLM 2026 workshop at EMNLP 2026) and card): does an answer contain claims its context does not support (example-F1: the answer is flagged or not).
- **RAGTruth:** does a retrieval-augmented answer contain claims its retrieved context does not support (example-F1).
- **MASSIVE, three unseen scenarios** (iot, transport, cooking; 52 languages): pick the intent and the scenario of a user command the model never saw in training (accuracy).
- **Super-NI task 1568** (ProPara): answer a multiple-choice question about how a process changes an entity (accuracy).

- **Zero-shot classification, 13 sets** (the GLiFormer paper's suite: SST-2, IMDB, AG News, Banking77, and more): pick the label of a text from label names only (mean macro-F1).
- **Decision Index 0.4.1** (board, 2026-10-28): 37 public decision benchmarks plus private tests; the Full score ranks on the private same-skill and new-domain tests (chance-corrected skill, 0 = guessing).
- **Router signal suite** (vllm-sr/router-signal-suite): the vLLM Semantic Router's public test of its request signals (harmful requests, jailbreaks, personal data, unsupported answers, fact-check need, modality, domain, feedback); each label comes from public corpora (AUC or accuracy).

The family



Model	Size	Base	Best for
vela3-fast	0.4B	UltraBERT-380M encoder	the default router; fastest: decisions + spans in one ~14 ms encoder pass, English, 32K
vela3-fast-vision	0.3B	UltraVBERT encoder	image decisions, image-grounded hallucination spans, region pointing (preview)

Model	Size	Base	Best for
vela3-lite	0.8B	d3-lite	small GPUs, low latency
vela3-mini	4B	d3-mini	balanced accuracy and speed
vela3-flash	9B	d3-flash	high-accuracy routing
vela3-turbo	27B	d3-turbo	the reference model

How it works

- **Frozen base.** `d3-lite@9c107031` runs unchanged; ordinary questions are its own answers.
- **Gated LoRA adapters.** Each switches on only for its own questions, and every answer names the one it used (`head_reason`): decision (router signals), NER (entity labels with definitions), relation, image (hallucination spans with images), span (evidence, PII, unsupported claims, toxic spans, names-only labels).
- **Spans as decisions.** The labels are presented as options; a span head scores every word against every label, and agreeing neighbours form a span with character offsets.
- **Label names in.** Labels are read by name at this size; an optional d3-mini writer (`definitions.labeldef_writer`) can write definitions first (OOD-7 64.4 with names only, 63.5 with the writer).
- **Specification.** Adds 154 MB of adapters and heads; image threshold 0.35; text or JSON

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state, images and video for decisions, images for hallucination spans. Full evaluation: EVALUATION.md.

Trade-offs

- **Unsupported claims:** hallucination spans 0.825 against 0.921 for LettuceDetect v2 qwen-2b, RAGTruth 0.675 against 0.850 for HDM-2 (3B) (reported).
- **General decisions are d3-lite's:** below GLiClass v4 Edge (0.4B) (full score 31.21 vs 26.16) and OneJev1.6-0.8B (full score 29.58 vs 26.16) on the Decision Index 0.4.1.
- **End-to-end relations are the weakest task:** 19.7 average on the four sets, against 21.3 for GLiFormer-large (reported).

Citation

```
@misc{vela3_lite,
  title        = {vela3-lite (Vela 3.0, 0.8B
  author       = {{KR Labs} and {vLLM Semant
  year         = {2026},
  howpublished = {\url{https://huggingface.c
}
```